

LOG SOURCES

**Windows**
Audit logs · Sysmon

**Linux**
rsyslog

**Beats agents**
Filebeat · Fluent Bit

**NIDS**
Suricata · Snort

**Firewalls**
Stormshield · Arkoon

**Web servers**
access / error

**Mail servers**
SMTP / IMAP

**File shares**
Samba · SharePoint

ETL — INGESTION

**Logstash #1**
normalization
syslog + JSON
→ RFC 5424

**Kafka**
streaming bus
retention buffer
topic: logs-raw

**ksqlDB**
categorization
splits per source
fw · nids · win ...

HOT PATH — DETECTION & RESPONSE · 2 MONTHS

**Logstash #2**
KV parse RFC 5424
GeoIP · CTI · assets

**Elasticsearch**
index per stream
ILM hot/warm · 2 mo

**Kibana**
dashboards
Sigma rule alerts

**TheHive**
SOC cases

**Cortex**
auto-response

COLD PATH — BIG DATA · 2 YEARS · 2 PB

**Hadoop HDFS**
legal retention 2 y
~2 PB

**Spark**
MapReduce batch
hunting at scale

LEGAL PATH

**Legal team stack**
independent consumer
chain of custody · forensics

TRANSVERSE

**IaC**
GitLab CI/CD
+ Ansible

**Monitoring**
Prometheus
+ Grafana

deploys & monitors every cluster

LEGEND

-  Ingestion
-  Hot path · detection
-  Cold path · big data
-  Legal path
-  Alerts & response
-  Deploy & monitor